How Likely Is a Cycle? The Spatial Version of Sen's §10.2 Question

Background note for the Ch 10 class. Sen's §10.2 asks for the probability of cyclical majorities under the "impartial culture" assumption that all orderings are equally likely, and reports Guilbaud's 8.77% limit for three alternatives (Table 10.1) and the rise towards 1 as the number of alternatives grows (Table 10.2). Sen is rightly unhappy with the assumption. Footnote 4:

> In this 'impartial culture', given a two-alternative choice between my being beheaded at dawn and my living on, the probability of my preferring either to the other will be exactly one-half. I protest.

The natural repair is to put voters and alternatives in a Euclidean space and let preference be given by distance. That is the spatial model, and it is the right setting for the question, because single-peakedness is the $k = 1$ case of it. This note reports what the literature has and what it does not, and gives simulation results for the general case.

The question

n voters and m alternatives are points drawn independently from some distribution on $\mathbb{R}^k$. Voter v prefers alternative A to B when $\|v - A\| < \|v - B\|$. What is the probability that the majority relation over the m alternatives has no Condorcet winner?

For $k = 1$ the answer is zero: Euclidean preferences on a line are single-peaked, so Black's theorem applies. Everything else is open territory.

What the literature has

The literature splits into two streams that do not quite meet in the middle.

Stream one takes every point in the space to be an alternative. This is the $m \to \infty$ limit. Plott (1967) gives the radial symmetry condition for a core to exist, and it is knife-edge. McKelvey (1976, 1979) shows that when symmetry fails, which is almost always, the top cycle is the entire space. Schofield (1978, 1983) adds genericity results. The verdict here is maximally pessimistic, and it is the result usually quoted.

Stream two takes a finite set of alternatives and random preference orderings. This is Sen's §10.2: Guilbaud (1952), Garman and Kamien (1968), Niemi and Weisberg (1968), later Gehrlein and Fishburn. It is not spatial at all, and it inherits exactly the objection in Sen's footnote 4. The most recent work here is Sauermann (arXiv:2203.13713, 2022), which shows that under impartial culture with $2k - 1$ voters and n alternatives the probability of a Condorcet winner is $\Theta(n^{-(k-1)/k})$, and which disproves Gehrlein's conjecture that impartial culture minimises that probability.

The bridge between them is the yolk. The yolk is the smallest ball meeting every median hyperplane; the concept is due to McKelvey (1986), building on Ferejohn, McKelvey and Packel (1984). The useful fact is a bound on win sets: Y beats X only if Y is at most $2r$ further from the yolk centre than X is, where r is the yolk radius. Feld, Grofman and Miller (1988) and Tovey ("The Almost Surely Shrinking Yolk", 1990 working paper, later in *Mathematical Social Sciences*) show the expected yolk radius goes to zero as the number of voters grows, for centred distributions. Tovey left the rate open; simulations reported in Miller's survey suggest an inverse square root law in n.

That bound does the work for our question. If the yolk shrinks and the alternatives are fixed in number, then the alternative nearest the yolk centre beats everything else as soon as $2r$ is smaller than the gaps between the alternatives' distances to the centre. So a Condorcet winner should exist with probability approaching one as n grows with m fixed. I have not found this stated as a theorem.

What I could not find. Nobody appears to have computed the probability for the general case with n, m and k all finite and both voters and alternatives drawn at random. The closest thing by title is Good and Tideman, "From individual to collective ordering through multidimensional attribute space", *Proceedings of the Royal Society of London A* 347 (1976): 371–385, which is the source of what Song (2021) calls the "normal spatial model", with ideal points drawn from a multivariate normal. I could not get at the paper and have not verified what it proves. **That is the one to check before relying on any of this.**

Simulation work exists but is scattered and mostly two-dimensional: Koehler (1990), Hug (1999), Bräuninger (2007), Miller (2007), and Tideman and Plassmann (2012), who report a cycle in 0.53% of simulated elections with 1,000 voters, falling towards zero as n grows. I have that figure at second hand from Song (2021) rather than from Tideman and Plassmann directly.

Simulation

Since the general numbers do not seem to be in print, here they are. Voters and alternatives are drawn independently from the same distribution, either standard normal or uniform on the unit cube in $\mathbb{R}^k$. 20,000 trials per cell. Entries are the proportion of trials with no Condorcet winner.

Points drawn from a standard normal

n	m	$k=1$	$k=2$	$k=3$	$k=5$	$k=10$
3	3	0.0000	0.0140	0.0173	0.0179	0.0215
3	5	0.0000	0.0459	0.0575	0.0688	0.0741
3	10	0.0000	0.1268	0.1510	0.1711	0.1716
11	3	0.0000	0.0100	0.0135	0.0156	0.0173
11	5	0.0000	0.0391	0.0452	0.0481	0.0519
11	10	0.0000	0.1152	0.1241	0.1224	0.1217

n	m	$k=1$	$k=2$	$k=3$	$k=5$	$k=10$
51	3	0.0000	0.0040	0.0043	0.0048	0.0056
51	5	0.0000	0.0137	0.0141	0.0152	0.0152
51	10	0.0000	0.0371	0.0398	0.0353	0.0355
201	3	0.0000	0.0008	0.0017	0.0016	0.0014
201	5	0.0000	0.0037	0.0036	0.0038	0.0041
201	10	0.0000	0.0112	0.0100	0.0098	0.0100
1001	3	0.0000	0.0001	0.0006	0.0002	0.0001
1001	5	0.0000	0.0010	0.0008	0.0004	0.0005
1001	10	0.0000	0.0024	0.0018	0.0019	0.0018

Points drawn uniformly from the unit cube

n	m	$k=1$	$k=2$	$k=3$	$k=5$	$k=10$
3	3	0.0000	0.0150	0.0231	0.0293	0.0358
3	5	0.0000	0.0526	0.0732	0.0917	0.1032
3	10	0.0000	0.1370	0.1749	0.2079	0.2233
11	3	0.0000	0.0173	0.0244	0.0278	0.0318
11	5	0.0000	0.0560	0.0746	0.0839	0.0881
11	10	0.0000	0.1325	0.1553	0.1729	0.1918
51	3	0.0000	0.0076	0.0101	0.0095	0.0106
51	5	0.0000	0.0190	0.0249	0.0309	0.0314
51	10	0.0000	0.0456	0.0515	0.0583	0.0657
201	3	0.0000	0.0022	0.0032	0.0039	0.0036
201	5	0.0000	0.0064	0.0073	0.0084	0.0102
201	10	0.0000	0.0114	0.0126	0.0179	0.0186
1001	3	0.0000	0.0004	0.0008	0.0006	0.0006
1001	5	0.0000	0.0011	0.0018	0.0026	0.0016
1001	10	0.0000	0.0027	0.0028	0.0032	0.0043

Letting the alternatives fill the space ($k=2$, uniform, 10,000 trials)

m	$n=3$	$n=11$	$n=51$
3	0.0141	0.0193	0.0067

m	$n = 3$	$n = 11$	$n = 51$
5	0.0507	0.0537	0.0212
10	0.1342	0.1316	0.0429
20	0.2627	0.2523	0.0844
50	0.4427	0.4804	0.1961
100	0.5677	0.6741	0.3462
200	0.6803	0.8347	0.5515
400	0.7545	0.9335	0.7515

Decay in the number of voters ($m = 3$, $k = 2$, 60,000 trials)

n	normal	uniform
11	0.01013	0.01818
21	0.00728	0.01325
51	0.00330	0.00755
101	0.00177	0.00425
201	0.00098	0.00203
501	0.00055	0.00123
1001	0.00022	0.00063
2001	0.00015	0.00018

What the numbers say

Dimension matters much less than the literature's tone suggests. The whole qualitative difference is between $k = 1$ and $k = 2$. After that the curve is nearly flat: going from three dimensions to ten barely moves the number, and in the normal case sometimes moves it down. If you had only read McKelvey you would expect high dimensionality to be catastrophic. With a finite set of alternatives it is not.

The number of alternatives is what drives cycles. Going from three alternatives to ten multiplies the probability by roughly eight, at every n and every k. The last table shows where this ends up: with the number of voters held fixed and the alternatives allowed to fill the space, the probability climbs towards one. That is the Plott and McKelvey result arriving as a limit rather than a theorem, and it shows what those results are really about. They are about the density of the alternative set, not about dimension.

The number of voters pushes hard the other way. With three alternatives and a thousand voters the probability is a few in ten thousand. Over the range where my estimates are reliable the decay is roughly $n^{-0.75}$, which sits between the $n^{-1/2}$ that the shrinking-yolk heuristic suggests

and n^{-1}. I cannot pin the exponent down: the large-n cells have very few events, so the estimates there are noisy, and the apparent steepening at the bottom of the last table should not be trusted. A plausible story for an exponent above $1/2$ is that a yolk-bound failure is necessary but not sufficient for a cycle.

The two limits pull in opposite directions, which is the useful thing to tell the class. Whether cycles are a real problem in the spatial model is not a question about dimension. It is a question about the ratio of alternatives to voters.

Caveats

The alternatives are drawn from the same distribution as the voters. Real candidates are not positioned at random and tend to cluster where voters are, which is Miller's argument for why cycles are rarer in practice than the geometry suggests. Drawing alternatives from a more dispersed distribution would push the numbers up.

The uniform and normal cases differ by roughly a factor of two throughout, so the distribution is doing real work and none of these figures should be quoted without it.

Simulation code: `tools/condorcet-cycles.py`.